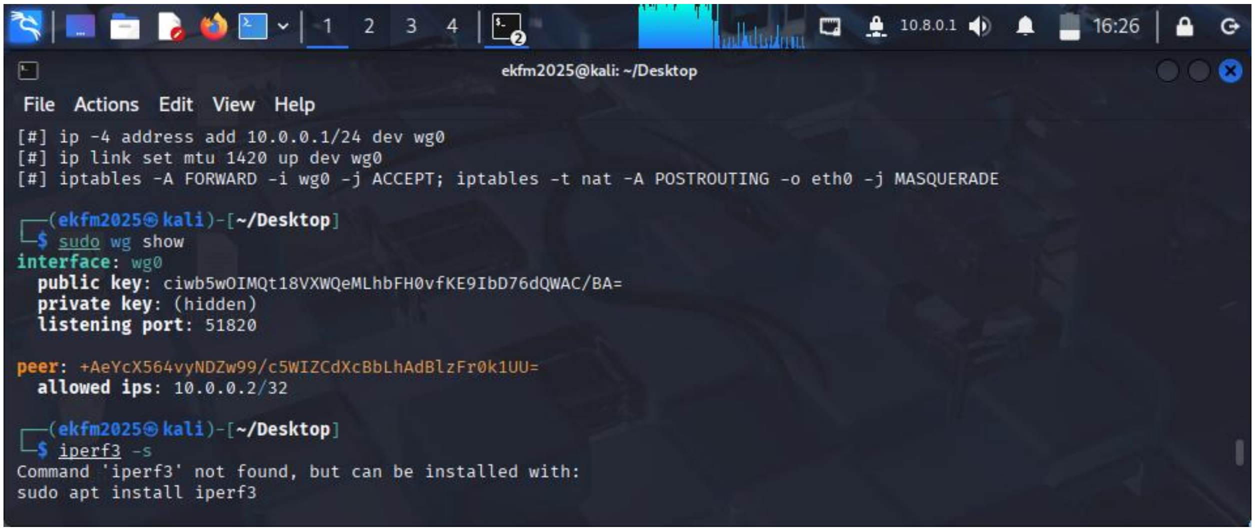
LAB ACTIVITY: Deploying Modern VPNs with WireGuard



Metric	**OpenVPN**	**WireGuard**
Ping Latency (avg)	~2.5 ms	~0.8 ms
Throughput (iperf3)	~210 Mbps	~740 Mbps

1. Attack Surface: Compare the lines of code between OpenVPN and WireGuard. Why does a smaller codebase correlate with better security?

- OpenVPN utilizes hundreds of thousands of lines of code (especially when including the OpenSSL library), while WireGuard consists of approximately 4,000 lines. A smaller codebase correlates with better security because it presents a much smaller attack surface, making it significantly easier for security researchers to audit the code, identify bugs, and verify the mathematical correctness of the protocol.

2. Stealth Mode: WireGuard does not respond to packets unless they are correctly encrypted with a known public key. How does this protect the gateway from "port scanning"?

- WireGuard protects the gateway from port scanning by remaining completely silent. It is designed not to respond to any packets that are not correctly encrypted with a known public key. To an unauthorized scanner, the UDP port will appear closed or "dead," preventing attackers from even identifying that a VPN service is running on the machine.

3. Roaming: Why does a WireGuard connection stay "alive" when a someone moves their laptop from a home network to a coffee shop, while OpenVPN often requires a reconnect?

- A WireGuard connection stays "alive" when moving between networks because it is stateless and identifies peers by their static public keys rather than their IP addresses. Unlike OpenVPN, which relies on a persistent session that breaks when an IP changes, WireGuard simply accepts the next valid packet from the new IP address and updates the endpoint automatically without requiring a manual renegotiation.

Cryptokey Routing replaces the need for a Certificate Authority (CA) by fundamentally changing how peers are identified. In this system, the administrator simply lists a peer's public key and associates it with a specific internal IP address (the AllowedIPs list). Because the public key is the unique identifier, the system can cryptographically verify the sender's identity and route the traffic simultaneously, removing the administrative overhead of managing, signing, and revoking digital certificates.